MEESHO PRAGATI AGENT

Agentic AI for Fair, Explainable Credit Access

Project Documentation & Technical Report

Theme: Building for Bharat with the Power of Agentic AI

Target Vertical: Meesho Finance (MPPL) & NBFC Lending Partners

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1. Executive Summary

Meesho Pragati Agent is an agentic AI system designed to solve a specific, verified business problem inside Meesho's own lending ecosystem: small-town sellers with no formal credit history are being rejected for working-capital loans by rigid, rule-based underwriting systems, despite having strong, verifiable performance records on the Meesho platform itself.

The system replaces a static reject/approve checklist with a three-layer decision pipeline: a trained machine learning model that computes a statistically grounded risk score from the seller's platform behaviour, a deterministic business-rules layer that converts that score into an auditable, RBI-compliant loan decision, and an orchestrating large language model (LLM) that interprets context, generates a human-readable explanation, and manages two-way WhatsApp communication with the seller in their own language.

Critically, the LLM never calculates the financial decision itself. This separation of concerns — model decides, rules finalize, LLM explains and acts — is the central design principle of this project, and is what allows the system to remain both intelligent and legally defensible under India's digital lending and data protection regulations.

This document presents the complete research, problem validation, solution design, technical architecture, machine learning methodology, business case, and risk mitigation strategy behind the project.

2. Research Background: Why This Topic Was Selected

Before finalizing this idea, three candidate directions were researched and evaluated against the hackathon theme ("Building for Bharat with the Power of Agentic AI") and against Meesho's actual, documented business priorities. The objective was to select a problem that was (a) genuinely under-served, (b) verifiably real rather than hypothetical, (c) technically feasible to prototype credibly within a hackathon timeframe, and (d) distinct from what the majority of competing teams were likely to submit.

2.1 Candidate Ideas Considered

Candidate A — Seller Catalog & Ad-Management Agent ("Vyapar-Mitra")

A WhatsApp-based multi-agent system to help rural sellers clean up product photos, generate SEO-optimized listings in vernacular languages, and autonomously manage advertising budgets. This idea directly targets Meesho's 0%-commission revenue model, which depends heavily on seller advertising income.

This was set aside as the primary submission because, on analysis, it represents the most predictable and over-subscribed interpretation of "Building for Bharat" — WhatsApp-first vernacular seller tools are the default brainstorm output for this theme, meaning it would face the highest competitive overlap of any candidate. It also carries higher live-demo risk, since it depends on real-time speech-to-text, image processing, and ad-bidding simulation all working correctly in front of judges simultaneously.

Candidate B — Fraud & Returns Forensics Agent ("Samadhaan")

A multimodal agentic system to autonomously investigate reverse-logistics fraud (e.g. empty-box returns, wrong-item returns) by cross-referencing unboxing video evidence, OCR-extracted tracking data, and courier-hub weight logs, issuing an automatic, explainable fraud ruling.

This was a strong second candidate, with high novelty and a direct EBITDA-protection narrative relevant to Meesho's pre-IPO cost discipline. It was set aside in favour of the final idea primarily because it depends on multimodal video/vision reasoning succeeding live, which introduces materially higher demo risk than a pure structured-data pipeline, and because the chosen idea offered a stronger, more specific tie to a named, currently-scaling Meesho business unit.

Candidate C — Agentic Underwriting for Seller Working Capital ("Kirana-Credit" / "Meesho Pragati Agent") — Selected

An agentic system that underwrites seller working-capital loans using alternative platform data instead of traditional credit scores. This was selected as the final idea.

2.2 Selection Rationale

The deciding factors, in order of weight given to each:

- Verified real-world grounding: this idea ties to a named, real, currently operating business unit — Meesho Payments Private Limited (MPPL), which runs the “Meesho Instant Cash” seller-lending product — rather than a hypothetical pain point.
- Lowest live-demo risk of the three candidates: the entire pipeline operates on structured, tabular JSON data. There is no dependency on real-time computer vision, video analysis, or speech recognition succeeding live in front of judges.
- Lowest expected competitive overlap: fintech/credit-underwriting agents are a substantially less obvious brainstorm output for an e-commerce hackathon than consumer-facing seller tools or fraud detection, based on typical idea-distribution patterns at corporate-sponsored hackathons.
- Strongest dual narrative: the idea serves both a humanitarian story (financial inclusion for unbanked entrepreneurs) and a hard commercial story (new high-margin revenue line, GMV protection) simultaneously — without one undermining the other.

2.3 Research Sources Consulted

The following research was conducted prior to finalizing the idea, to validate that the underlying problem is real rather than assumed:

- Verified that Meesho Payments Private Limited (MPPL) operates as a Lending Service Provider (LSP), originating loan applications and handling customer servicing while partner NBFCs supply the underlying capital — confirming the architectural role this project's agent would occupy.
- Verified that MPPL received a capital infusion of approximately ₹100 crore via a rights issue specifically to scale its lending-service-provider operations, and that its revenue grew sharply year-on-year (from roughly ₹0.20 crore to ₹11.05 crore) while it remained loss-making (a net loss of approximately ₹24.72 crore) in the same period — establishing that this is an actively scaling, currently underperforming business unit, not a speculative target.
- Verified, via a Government of India Press Information Bureau (PIB) release, that AI-driven alternative-data lending models are estimated to be capable of unlocking USD 130–170 billion of India's MSME credit gap, an estimate distinct from (and narrower than) broader total-credit-gap figures, giving the project a defensible macro-level justification.
- Verified that India's MSME sector remains predominantly cash-based, meaning conventional lenders frequently cannot verify a small business's true cash flow in the absence of GST filings or bank-verified income statements — directly substantiating the “thin-file seller” problem framing.
- Verified that the cost of manually underwriting a small-ticket loan is close to the cost of underwriting a large one, which structurally disincentivizes lenders from serving small borrowers profitably under manual review — this finding directly informed the project's two-tier ML-first, LLM-for-edge-cases architecture, described in Section 10.

- Reviewed AWS and Twilio/Meta published pricing documentation for GPU inference hosting and WhatsApp Business API messaging in the Indian market, to ground the cost estimates in Section 21 in real, current figures rather than assumptions.

3. Problem Statement

Millions of small-town entrepreneurs run genuinely successful businesses on Meesho while operating almost entirely in the informal, cash-based economy. They have no CIBIL score, no formal bank loan history, and typically no collateral to offer. When seasonal demand spikes occur — wedding season, Diwali, regional festivals — these sellers frequently lack the working capital to purchase additional inventory in time to capture that demand.

Meesho already operates a seller-lending product, Meesho Instant Cash, through its finance subsidiary MPPL, in partnership with third-party NBFCs. However, the underwriting logic behind this product today is reported to rely on rigid, rule-based checks inherited from traditional lending practice — fixed thresholds such as a minimum CIBIL score or a minimum account age. A seller who fails even one such threshold, for any reason — including a temporary, explainable disruption such as regional flooding — is auto-rejected, with no explanation given and no path offered to reapply.

The consequence is twofold. The seller loses trust in the platform and is unable to grow their business at the moment they most need to. Meesho loses the Gross Merchandise Value (GMV), the advertising revenue, and the lending commission that a well-capitalized, growing seller would otherwise generate — at the exact moment its own finance subsidiary needs to become profitable.

3.1 Why This Is Solvable Specifically by Meesho

The critical insight behind this project is that Meesho already possesses everything required to assess a seller's reliability — it simply isn't using that data for credit decisions today. Sales velocity, return-to-origin (RTO) rate, dispatch SLA compliance, advertising ROI, and customer ratings are all continuously generated by normal platform activity. A seller who is invisible to a traditional bank may be extremely well-characterized inside Meesho's own systems.

4. Existing Solutions & Industry Landscape

4.1 Meesho's Current Approach

Meesho Instant Cash currently operates via partner NBFCs applying conventional, rule-based eligibility checks — for example, a minimum CIBIL score, a minimum number of months active on the platform, or a minimum monthly revenue threshold. If a seller fails any single rule, the application is rejected outright, with a generic decline message and no further recourse offered within that cycle.

4.2 Broader Industry Approaches

Other lending-service-provider and NBFC-partnership models in Indian fintech (comparable in structure to programs run by Flipkart and Amazon India for their own seller bases) follow a similar pattern: alternative-data signals are sometimes used as inputs, but the decisioning logic itself typically remains a static scorecard or a fixed rules engine rather than a reasoning system capable of incorporating context.

4.3 Limitations of the Current Approach

- No context-awareness: a temporary, explainable dip in performance (e.g. due to local disruption) is treated identically to genuine financial distress.
- No explainability: a rejected seller receives no reason and no actionable guidance.
- No follow-up mechanism: a seller who improves their metrics after rejection has no automatic path back to reconsideration — they must proactively reapply, with no visibility into whether they now qualify.
- High per-application manual-review cost for borderline cases, which discourages lenders from investing reviewer time in small-ticket loans — a documented, structural issue in small-business lending generally, not specific to Meesho.

5. Gap Identified

The core gap is the absence of a reasoning layer between Meesho's rich platform data and the NBFC partner's lending decision. Today, that gap is filled by nothing — the NBFC simply applies its standard external scorecard, blind to everything Meesho already knows about the seller.

This produces three concrete, addressable sub-gaps:

1. Data gap: a seller's real operational performance on Meesho is not being translated into a format a lender can act on.
2. Reasoning gap: even where alternative data is considered, static rules cannot distinguish a temporary, explainable anomaly from genuine credit risk.
3. Communication gap: rejected sellers are given no explanation and no improvement path, causing avoidable trust erosion and seller churn.

Meesho Pragati Agent is designed to close exactly these three gaps — and no more. It does not attempt to replace the NBFC's regulatory role, issue binding credit decisions independently, or operate without human oversight on larger loans; its scope is deliberately bounded to closing the data, reasoning, and communication gap inside Meesho's existing lending flow.

6. Our Solution: Detailed Description

Meesho Pragati Agent sits as an intelligent layer between Meesho's seller data and the NBFC partner's final lending decision. It does not replace the NBFC's legal authority to approve or deny credit; it makes the data reaching that decision dramatically richer, faster to assess, and accompanied by a clear, auditable explanation.

6.1 Core Design Principle

The single most important architectural decision in this project is the strict separation between three distinct layers of intelligence, each with a clearly bounded responsibility:

- A trained machine learning model computes the risk score and base loan eligibility from historical, structured seller data, using standard statistical/ML methods (Section 10).
- A deterministic rules engine takes that score and applies fixed, auditable business logic to set the final loan number — ensuring the actual financial decision is reproducible and compliant with RBI digital lending norms (Section 8).
- An orchestrating LLM-based agent interprets the model's output, incorporates unstructured context (for example, a local-disruption signal), generates a plain-language explanation, and manages the conversational WhatsApp interaction with the seller (Section 11).

This means the LLM never sets a loan amount and never makes the underlying financial decision — it explains, contextualizes, and communicates a decision that was computed deterministically. This is the project's direct answer to the most predictable and serious objection any reviewer could raise (see Section 25).

6.2 What the Seller Experiences

From a seller's point of view, the experience is simple: they request a loan (or are proactively offered one based on a detected demand signal), and within seconds they receive a WhatsApp message in their own language that either confirms approval and disbursal, or explains — specifically and constructively — what they need to improve and by how much to qualify.

6.3 What the NBFC / Meesho Finance Team Experiences

On the lending-partner side, every decision the system produces is accompanied by a structured “Reasoning Trail”: a short, plain-language record of which signals were considered and how they led to the outcome. This is designed specifically to satisfy audit and compliance requirements under RBI digital lending guidelines, which require credit decisions to be explainable and traceable.

7. System Workflow

7.1 The Five-Step Core Pipeline

1. Gather — the system pulls the seller's trailing six months of platform activity: sales velocity, return-to-origin (RTO) rate, dispatch SLA compliance, advertising ROI, and customer rating trend.
2. Score — a trained ML model converts this feature set into a statistically grounded risk score and a base loan eligibility figure (Section 10).
3. Reason — the orchestrating LLM agent interprets the score in context: it checks for unstructured signals (e.g. a regional disruption) that might explain an anomalous metric, and determines whether the case is a straightforward approval, a straightforward decline, or a borderline case warranting deeper reasoning.
4. Decide — a deterministic rules layer converts the (possibly context-adjusted) score into the final, auditable loan limit and terms.
5. Act — the agent disburses (subject to NBFC KYC/liquidity checks) or sends a personalized, vernacular coaching message via WhatsApp, and schedules an automatic weekly re-check for borderline/declined sellers.

7.2 Illustrative Scenarios

Scenario 1 — High-Speed Acceptance

A seller requests ₹30,000 for festive stock. The model finds no formal credit history but an RTO rate of 4% and dispatch compliance of 98% — both well within a low-risk profile. The system approves automatically and disburses, subject to standard KYC, with a WhatsApp confirmation in the seller's language.

Scenario 2 — Milestone Coaching

A seller applies with a 14% order-cancellation rate, well above the 5% safe threshold. Rather than a flat decline, the system calculates the precise improvement needed, messages the seller with a specific 30-day target, and schedules an automatic weekly recheck — proactively notifying the seller the moment they qualify.

Scenario 3 — Context-Aware Guardrail

A consistently strong seller shows a sudden 60% sales drop over two weeks. A static rule-based system would auto-flag this as high risk. The agent instead checks for an explanatory signal — for example, regional flooding disrupting local transport — and, finding the underlying business healthy, can approve a short-term relief loan with adjusted repayment terms rather than auto-rejecting a fundamentally sound borrower.

8. Technical Architecture

The architecture is organized into four layers: the seller-facing interface layer, the orchestration layer, the intelligence layer (split between the ML scoring service and the LLM reasoning service), and the output/audit layer.

8.1 Layer-by-Layer Breakdown

Layer 1 — Interface

- WhatsApp (via Twilio API) for all seller-facing communication — inbound loan requests, outbound approvals, coaching messages, and status updates, delivered in the seller's preferred language.
- A React.js web dashboard for the NBFC / Meesho Finance admin team, displaying each seller's profile, risk score, and full Reasoning Trail for audit purposes.

Layer 2 — Orchestration

- A Node.js + Express.js backend service acts as the central orchestrator: it receives triggers (a seller's loan request, or a proactive demand-spike signal), sequences calls to the ML scoring service and the LLM reasoning service, and writes the final decision and Reasoning Trail to the database.
- LangChain.js manages the agentic control flow — tool-calling for data retrieval, routing between the deterministic rules engine and the LLM reasoning step, and managing multi-turn WhatsApp conversation state.

Layer 3 — Intelligence (split, by design)

- ML Scoring Service: a Python microservice running a trained XGBoost gradient-boosted model, which computes the risk score and base eligibility figure directly from structured seller data (see Section 10 for full methodology).
- LLM Reasoning Service: a self-hosted open-source LLM (Llama 3.1 8B or Qwen 2.5 7B, served via Ollama or vLLM), used exclusively for: interpreting unstructured context, generating the plain-language Reasoning Trail, and drafting the vernacular WhatsApp message. The LLM output never overrides the ML model's score or the rules engine's final number.
- BGE-Small / MiniLM embeddings support similarity lookups against historical seller patterns and disruption-signal text (e.g. matching a seller's stated reason against known regional-disruption reports).

Layer 4 — Output & Audit

- MongoDB stores seller transaction histories, decision records, and full Reasoning Trails for every case processed.
- AWS S3 retains long-term audit logs to satisfy regulatory record-keeping requirements.

8.2 Data Flow Summary

Seller / WhatsApp trigger → Node.js/Express orchestrator → [parallel: MongoDB data pull + XGBoost risk scoring] → LangChain.js reasoning step (LLM interprets score + context) → deterministic rules engine (locks final loan number) → decision written to MongoDB/S3 (audit trail) → NBFC admin dashboard (React) for audit visibility → Twilio WhatsApp outreach back to the seller.

The single design detail worth emphasizing again here: the arrow from the ML scoring box to the LLM reasoning box flows one way for the purposes of explanation generation, but the arrow that actually sets the loan number flows from the ML model directly into the deterministic rules engine, bypassing the LLM entirely. This is the architectural guarantee that the financial decision itself remains fully deterministic and auditable.

9. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	NBFC admin dashboard; seller-facing web views
Backend / Orchestration	Node.js + Express.js	Central orchestrator; REST APIs; webhook handling
Agentic Framework	LangChain.js	Task routing, tool-calling, multi-turn reasoning chain
ML Risk Model	Python microservice (XGBoost / scikit-learn)	Trained risk-scoring model; deterministic eligibility math
LLM (Open Source)	Llama 3.1 8B / Qwen 2.5 7B (Ollama / vLLM)	Reasoning, context interpretation, explanation, conversation
Embeddings	BGE-Small / MiniLM	Similarity matching for historical patterns & context signals
Database	MongoDB	Seller transaction & alternative-data store
Messaging	Twilio WhatsApp API	Vernacular seller outreach (approvals, coaching, status)
Infrastructure	Docker, AWS EC2 (GPU instance for LLM), AWS S3	Containerized deployment; inference hosting; audit log storage

> *Design rationale: open-source, self-hosted models keep sensitive financial data inside Meesho's own infrastructure (rather than sending it to a third-party API) and make nightly scoring of Meesho's full seller base economically viable, since cost is fixed GPU-hosting cost rather than a per-call API fee that scales with seller count.*

10. Machine Learning Methodology

This section specifies the actual machine learning approach underlying the risk-scoring layer, since this is the component that performs the real financial decisioning and therefore needs to be the most rigorously specified part of the system.

10.1 Why a Trained Model, Not an LLM, Performs the Scoring

Two independent reasons drove this decision, both confirmed during research:

1. Regulatory: RBI digital lending guidance requires credit decisions to be deterministic and auditable. A large language model is non-deterministic by construction — it can produce different outputs for the same input — and therefore cannot be the system that calculates a financial number a regulator might later need to reproduce exactly.
2. Computational: seller performance data (sales volume, RTO rate, account age) is structured, tabular data. Running a full LLM over millions of rows of structured JSON is computationally wasteful compared to a model purpose-built for tabular data, which costs a fraction of a cent and runs in milliseconds per prediction.

10.2 Algorithm Selection: Gradient-Boosted Decision Trees (XGBoost)

XGBoost was selected over alternative approaches for the following reasons:

- Tabular performance: gradient-boosted trees are widely regarded as the strongest-performing class of model for structured, tabular financial and behavioural data — the exact data type this project works with.
- Native explainability support: tree-based models support feature-importance extraction and SHAP (SHapley Additive exPlanations) value computation, allowing the system to state precisely which features drove a given score — directly supporting the Reasoning Trail requirement.
- Handles missing/sparse data well: many “thin-file” sellers will have incomplete historical data (e.g. a newly onboarded seller); tree-based models degrade gracefully in this scenario, unlike many linear or deep-learning approaches.
- Low compute cost at inference time: a trained XGBoost model can score a seller in single-digit milliseconds on CPU, supporting nightly batch scoring of the full seller base at minimal infrastructure cost.

10.3 Feature Set (Model Inputs)

Feature category	Example features
Sales performance	Trailing 6-month sales velocity; month-over-month growth rate; seasonal sales pattern

Feature category	Example features
Fulfilment reliability	Return-to-Origin (RTO) rate; order cancellation rate; dispatch SLA compliance %
Customer trust signals	Average seller rating; rating trend (improving / declining); complaint rate
Platform engagement	Ad-spend ROI; account age on platform; catalog size and update frequency
Risk history	Prior loan repayment history (if any); previous default/late-payment flags

10.4 Target Variable & Training Approach

The model is framed as a supervised learning problem with two linked outputs: a binary risk classification (low / medium / high risk) and a regression output estimating a safe maximum loan ceiling. In production, this would be trained on Meesho's historical loan-performance data — i.e. past loans where the eventual repayment outcome (on-time, late, defaulted) is known, allowing the model to learn which platform-behaviour patterns correlate with real repayment outcomes.

For prototype/demo purposes, a synthetic dataset is generated to approximate realistic seller behaviour distributions: a set of simulated seller profiles (suggested: 500–1,000 synthetic records) with feature values drawn from plausible ranges (e.g. RTO rate 2–25%, account age 1–60 months, rating 3.0–5.0), with a labelled outcome generated by a rule-based ground-truth function during dataset construction, which the model is then trained to approximate and generalize beyond — explicitly disclosed as synthetic data standing in for real historical loan-performance data Meesho would use in production.

10.5 Explainability Layer

On top of the trained XGBoost model, SHAP value computation is used to generate a ranked list of which features contributed most to a given seller's score, in which direction. This ranked list is what the LLM reasoning layer (Section 11) translates into the plain-language Reasoning Trail shown to both the seller and the NBFC auditor — ensuring the explanation is grounded in the model's actual computed feature attributions, not an LLM-generated guess at why a decision was made.

10.6 Deterministic Rules Layer (Post-Model)

The model's raw score is not used directly as the final loan decision. A deterministic rules layer applies fixed business logic on top of the model output — for example, hard caps on maximum loan size by seller tenure, mandatory human review above a fixed threshold (Section 25), and rounding/tiering rules for final disbursement amounts. This final layer is pure, auditable arithmetic and conditional logic, with no model or LLM involvement, which is what allows the system to state with confidence that the actual financial number is fully reproducible and explainable to a regulator.

11. Agentic AI Layer — Detailed Working

This section specifies precisely what the LLM-based agentic layer does, and — equally importantly — what it deliberately does not do, since this distinction is the project's core safety and compliance argument.

11.1 Definition of “Agentic” as Applied Here

An agent, in this system, is defined as a component that can autonomously sequence multiple steps — gathering information, reasoning over it, and taking a real-world action — without a human manually triggering each step. This is distinct from a single-shot model call that returns one prediction and stops.

11.2 What the Agent Is Responsible For

1. Tool-calling / data orchestration: the agent (via LangChain.js) decides which data sources to query for a given seller — platform transaction history, and, for borderline cases, unstructured context sources (e.g. a seller's own chat message explaining a disruption).
2. Context interpretation: for cases the ML model flags as borderline or anomalous, the agent reasons over unstructured signals (for example, recognizing that a sudden sales drop coincides with a reported regional flooding event) to determine whether an anomaly has a benign explanation.
3. Explanation generation: the agent translates the ML model's SHAP-based feature attributions into a clear, plain-language Reasoning Trail, in the appropriate register for two different audiences — a seller-facing version (simple, encouraging, actionable) and an auditor-facing version (precise, structured, traceable).
4. Conversational action: the agent manages the actual WhatsApp interaction — sending the initial outcome message in the seller's language, handling simple reply flows (e.g. “Reply YES to accept”), and triggering the scheduled weekly re-check loop for borderline/declined sellers.

11.3 What the Agent Is Explicitly Not Responsible For

- It does not calculate the risk score — that is the trained XGBoost model's exclusive responsibility (Section 10).
- It does not set the final loan amount — that is the deterministic rules engine's exclusive responsibility (Section 8.2).
- It does not issue a final, binding credit approval — outputs are framed as pre-qualified estimates subject to the NBFC's KYC and liquidity checks, and loans above a defined threshold are automatically routed to a human underwriter (Section 25).

11.4 Why an Agent Is Necessary, Not Merely an LLM Call

A single LLM call could, in principle, generate an explanation for a single decision. What it cannot do on its own is: decide which data to fetch for a specific seller, recognize when a case needs deeper unstructured-context investigation versus a straightforward pass-through, manage a multi-turn conversation that may span days (coaching → weekly recheck → eventual qualification), and trigger that recheck autonomously without a human re-initiating it each time. These are precisely the properties — autonomous multi-step sequencing, tool use, and persistent follow-through — that define an agent rather than a static model call, and they are the reason this problem specifically benefits from agentic AI rather than a simpler ML-only or rules-only system.

12. How the Solution Solves the Problem

Problem (Section 3)	How the solution addresses it
No formal credit history	ML model scores sellers using platform behaviour data instead of CIBIL score
Rigid, single-rule rejection	Model + rules layer evaluate the full feature set together, not one threshold in isolation
No context for anomalies	Agent checks for explanatory context (e.g. local disruption) before treating a dip as risk
No explanation given on rejection	Every decision generates a Reasoning Trail, shown to the seller in plain language
No path to reapply	Coaching messages give a specific target; automatic weekly recheck re-evaluates without manual reapplication
Manual review too costly for small loans	ML handles the high-volume clear-cut cases automatically; LLM agent reasoning is reserved for the smaller share of borderline cases

13. Validation Strategy

13.1 Prototype-Stage Validation

- Synthetic seller dataset (Section 10.4) used to validate that the model produces sensible, monotonic risk scores — e.g. confirming that, holding other features constant, increasing RTO rate consistently increases predicted risk.
- Manual review of a sample of generated Reasoning Trails to confirm the LLM's plain-language explanation accurately reflects the model's actual SHAP feature attributions, rather than generating a plausible-sounding but disconnected explanation.
- End-to-end workflow test across all three demo scenarios (Section 7.2: acceptance, coaching, context-aware guardrail) to confirm the full pipeline — data gather → score → reason → decide → act — completes correctly for each case type.

13.2 Production-Stage Validation (Recommended Path)

- Backtesting the trained model against Meesho's actual historical loan-performance data (where available), comparing model-predicted risk tiers against real repayment outcomes.
- A/B testing the agent-assisted underwriting flow against the existing rule-based flow on a held-out segment of real applicants, measuring approval rate, default rate, and seller satisfaction/retention as primary outcome metrics.
- Periodic model re-validation and drift monitoring, given that seller behaviour patterns (e.g. seasonal effects) will shift over time.

14. Potential Impact

The potential impact spans both the seller side and Meesho's own business metrics, and the two are directly linked:

- For sellers: working capital becomes accessible based on demonstrated platform performance rather than an absent formal credit history, with a guided path to eligibility even for those who are initially declined.
- For Meesho: a higher loan approval-conversion rate at MPPL directly addresses the documented loss-making position of that business unit; better-capitalized sellers are expected to stock more inventory, list more products, and spend more confidently on Meesho Ads, with downstream effects on GMV, ad revenue, and seller retention.

This dual structure — financial inclusion and commercial upside reinforcing each other rather than trading off — is the central impact argument of this project.

15. Feasibility & Viability

15.1 Technical Feasibility

High. The system operates entirely on structured, tabular data, avoiding the higher technical risk associated with computer vision or video-based pipelines. Every component (XGBoost, LangChain.js, Twilio, MongoDB) is mature, well-documented technology with low integration risk.

15.2 Business Viability

High. Lending is a structurally high-margin business once a reliable underwriting signal exists — the limiting factor for MPPL today, per available research, is its ability to scale this lending-service-provider function profitably and reliant on NBFC partnerships, which this project directly targets.

15.3 Regulatory Viability

Conditional on the architectural separation described in Section 8.2 and Section 11.3 being strictly maintained — i.e. the LLM never calculates the financial decision, every decision is explainable and reproducible, and human review is enforced above a defined loan-size threshold. This is addressed in detail in Section 25.

16. Technical Excellence

- Clear separation of concerns across three distinct intelligence layers (ML scoring, deterministic rules, LLM orchestration), rather than collapsing all decisioning into a single model call.
- Explainability is built in at the model level (SHAP feature attribution) rather than bolted on afterward as an LLM-generated narrative disconnected from the actual decision logic.
- Cost-aware architecture: high-compute LLM reasoning is reserved for borderline cases rather than applied uniformly to the entire seller base, keeping inference cost proportional to where it adds genuine value.
- Self-hosted, open-source LLM and embedding stack (Llama 3.1 8B / Qwen 2.5 7B, BGE-Small) keeps sensitive financial data inside Meesho's own infrastructure rather than routing it through a third-party API.
- Designed-in compliance: deterministic decisioning, full audit trails, and mandatory human review above a defined threshold are architectural features, not afterthoughts added in response to anticipated objections.

17. Innovation & Creativity

The majority of e-commerce hackathon submissions for a “Building for Bharat” theme default to consumer-facing seller tools — catalog assistants, ad-management bots, vernacular chat support. This project instead pivots to a B2B fintech underwriting agent, applying agentic AI to exactly the high-margin financial-services layer that a platform like Meesho needs to scale profitably ahead of an IPO.

The specific innovation is not “using AI for lending” in the abstract — alternative-data lending itself is an established direction in Indian fintech — but rather the explicit, principled separation between a deterministic scoring/decisioning core and an agentic reasoning/communication layer wrapped around it. This directly answers the most common objection to AI in regulated lending (“you can't let an LLM decide a loan”) by design, rather than needing to argue around it after the fact.

18. Alignment with Theme

Theme: “Building for Bharat with the Power of Agentic AI.”

18.1 “Building for Bharat”

This is interpreted not as a UX/localization exercise (vernacular interface, WhatsApp delivery — both of which this project also includes) but as removing a structural barrier: the absence of formal credit infrastructure that locks capable small-town entrepreneurs out of growth capital, regardless of how well their business is actually performing.

18.2 “Power of Agentic AI”

The agentic component is not a feature added to an existing workflow — it is the mechanism that makes the solution possible at all. A static rule or a single ML score cannot reason about context, generate a tailored explanation, or autonomously follow up over time. Only a system capable of multi-step, autonomous reasoning and action can replace the function of a human credit officer across millions of sellers simultaneously. Remove the agentic layer, and the system reduces to the same rigid, unexplained rejection problem Meesho's lending product has today.

19. Business Impact & Business Plan

19.1 Revenue Mechanisms

Mechanism	Description
Lending referral commission	Meesho/MPPL earns a commission on every loan successfully originated and disbursed through the NBFC partnership
GMV uplift	Better-capitalized sellers stock more inventory and fulfil higher order volumes, directly increasing platform GMV
Ad revenue uplift	Sellers with available working capital are more likely to invest confidently in Meesho Ads
Seller retention	Sellers who receive capital support (or a clear coaching path) during a growth or crisis moment are less likely to churn off the platform

19.2 Phased Rollout Plan (Recommended)

1. Phase 1 — Shadow mode: run the model and agent alongside the existing rule-based system without acting on its output, to validate score quality against real outcomes before any live decisions are made.
2. Phase 2 — Limited pilot: deploy live for a small, defined seller cohort and loan-size cap, with full human review on every decision.
3. Phase 3 — Scaled deployment: expand the automated-decision threshold gradually as validated accuracy and compliance performance are demonstrated, always retaining mandatory human review above the defined high-value threshold (Section 25).

20. Impact on Bharat

- Converts demonstrated hard work and platform performance into bankable trust for sellers who have been structurally invisible to formal credit their entire business life.
- Enables proactive stocking ahead of festive and seasonal demand, rather than forcing sellers to cap orders or turn away sales due to lack of capital.
- Replaces a single, unexplained rejection with a specific, actionable improvement path — turning a demoralizing dead end into a guided route to eventual qualification.
- Aligns directly with the Government of India's own framing of AI-driven alternative-data lending as a mechanism to close a significant share of the national MSME credit gap, rather than positioning this as a Meesho-specific or hackathon-specific framing alone.

21. Cost of Implementation

All figures below are grounded in current, published pricing (AWS and Twilio/Meta rate cards) rather than estimates, to ensure this section can withstand direct scrutiny.

Cost component	Basis	Approximate cost
Self-hosted LLM inference (GPU)	AWS g4dn.xlarge on-demand instance (sufficient for an 8B-parameter open-source model)	~$0.526/hour (~$378/month), fixed regardless of seller volume
ML risk model (training + inference)	XGBoost on standard CPU instance; training is one-time/periodic, inference is near-zero marginal cost	Negligible per-prediction cost; standard compute instance pricing
WhatsApp outreach (Twilio + Meta)	India is among the lowest-cost WhatsApp markets globally; Twilio adds a flat per-message platform fee on top of Meta's base rate	~₹1–2 per message sent (varies by message category)
Database & storage	MongoDB hosting + AWS S3 for audit logs	Standard managed-service pricing, scales with data volume

> *Because the LLM reasoning layer is invoked only for borderline cases rather than the full seller base, GPU hosting cost remains roughly fixed (~$378/month) even as seller volume scales — the only cost component that scales directly with volume is WhatsApp messaging, which is itself inexpensive in the Indian market. This compares favourably against the cost of manual human underwriting review, where small-ticket loans are documented to cost lenders nearly as much to review as large ones, discouraging profitable service of small borrowers today.*

22. Technical Functionalities

- Automated structured-data ingestion from seller transaction history (sales, RTO, dispatch SLA, ratings, ad ROI).
- Trained gradient-boosted (XGBoost) risk-scoring model with SHAP-based feature attribution.
- Deterministic, auditable rules engine for final loan-limit calculation.
- LLM-based agentic orchestration layer (LangChain.js) for context reasoning, explanation generation, and conversational management.
- Self-hosted open-source LLM inference (Llama 3.1 8B / Qwen 2.5 7B via Ollama/vLLM) and embedding-based similarity matching (BGE-Small/MiniLM).
- Automated, scheduled weekly re-evaluation loop for borderline/declined sellers.
- WhatsApp Business API integration (via Twilio) for vernacular, bidirectional seller communication.
- Full decision audit-trail logging (MongoDB + S3) for regulatory and internal compliance review.
- React.js-based admin dashboard for NBFC/Meesho Finance audit and oversight.

23. Non-Technical Functionalities

- Plain-language, vernacular explanation of every loan decision — approval or decline — delivered directly to the seller.
- Specific, actionable coaching guidance for sellers who do not currently qualify, rather than a generic rejection.
- Proactive, unprompted outreach to sellers when a favourable lending opportunity (e.g. a detected demand spike) is identified.
- Transparent disclosure to sellers that any AI-generated offer is a pre-qualified estimate, subject to final NBFC checks — to manage expectations honestly (Section 25).
- An auditable, human-readable explanation available to NBFC compliance staff for every decision made, supporting regulatory review without requiring technical interpretation of model internals.

24. Stakeholders & Users Involved

Stakeholder	Role / Interaction with the system
Meesho seller	Requests or is proactively offered a loan; receives decisions and coaching via WhatsApp
Meesho Finance / MPPL	Owns the lending-service-provider relationship; uses the system to improve loan origination quality
NBFC lending partner	Supplies underlying capital; relies on the system's Reasoning Trail for compliant, auditable decisioning
Human underwriter (compliance role)	Reviews and approves loans above the defined high-value threshold; reviews flagged edge cases
Regulatory / compliance function	Audits decision trails for RBI digital lending and DPDP Act compliance

25. Known Limitations & Mitigations

This section addresses, directly and without minimizing them, the most significant risks a technically informed reviewer would raise against this system. Each is paired with the specific architectural decision made to address it.

25.1 Regulatory Risk: RBI Digital Lending Guidelines

Risk: RBI's digital lending guidance requires that credit decisions be deterministic and auditable. A large language model is non-deterministic and cannot legally be the component that calculates a financial decision, since its output cannot be guaranteed reproducible for audit purposes.

Mitigation: the LLM is architecturally restricted to orchestration, context interpretation, and communication. The actual risk score is computed by a deterministic, trained model (Section 10), and the final loan number is set by a fixed rules engine (Section 8.2) with no LLM involvement. The system can state, accurately, that the financial decision is fully reproducible independent of any LLM call.

25.2 Computational Efficiency: Why Not Run an LLM Over All Seller Data

Risk: structured, tabular seller data (sales figures, RTO rate) is not the data type LLMs are designed for, and running an LLM over millions of such records would be slow and computationally wasteful compared to a model purpose-built for tabular inputs.

Mitigation: the trained XGBoost model handles the large majority of clear-cut approval/decline cases directly. The LLM agent is invoked specifically for the smaller share of borderline cases that benefit from conversational reasoning and unstructured-context interpretation — concentrating compute cost where it adds genuine decisioning value rather than applying it uniformly.

25.3 Moral Hazard: Risk of False Promises to Sellers

Risk: if the agent autonomously tells a seller “improve metric X and you will receive ₹Y,” and the NBFC's broader credit policy subsequently shifts, the seller may be denied despite meeting the agent's stated target — causing a direct trust breach attributable to Meesho.

Mitigation: the agent is explicitly not authorized to issue binding financial commitments. Every outbound message is framed as a pre-qualified estimate, with a disclosure that final disbursement remains subject to the NBFC's KYC and liquidity checks. Additionally, any loan above a defined threshold (for example, ₹1 lakh) is automatically routed to a human underwriter for one-click final review before any commitment is communicated to the seller.

25.4 Data Privacy: India's Digital Personal Data Protection (DPDP) Act

Risk: aggregating a seller's ad-spend, cancellation, and geographic data to share with an NBFC partner requires explicit, informed consent under the DPDP Act; doing so without proper consent exposes Meesho to significant regulatory liability.

Mitigation: the alternative-data underwriting workflow is triggered only after the seller explicitly opts in via a clearly presented dashboard control, which generates an auditable consent token stored alongside the resulting decision record — ensuring the data-access event itself is traceable and consent-backed.

25.5 Model and Dataset Limitations (Prototype Stage)

The ML model described in Section 10 is, at the prototype stage, trained on a synthetic dataset constructed to approximate plausible seller-behaviour distributions, since real historical loan-performance data was not available outside Meesho's own systems. This is disclosed explicitly rather than implied to be production data. Before any real deployment, the model would require retraining and backtesting against Meesho's actual historical loan outcomes, as outlined in the production-stage validation plan (Section 13.2).

26. Conclusion

Meesho Pragati Agent addresses a verified, currently unresolved gap inside a real, actively scaling, and currently loss-making Meesho business unit. It does so by combining a deterministic, auditable machine learning core with an agentic reasoning and communication layer built specifically to satisfy the regulatory, computational, and trust-related constraints that any credible lending-AI system must address.

The project's central design principle — that intelligence and decisioning are deliberately separated, with the LLM agent restricted to orchestration, explanation, and communication rather than financial calculation — is what allows this system to be both genuinely agentic in capability and legally defensible in deployment. This is not AI applied to a problem for its own sake; the autonomous, multi-step reasoning of an agent is the specific mechanism that allows a static rejection process to become an explainable, context-aware, and ultimately more inclusive one, at a scale no purely manual process could match.

> *Building for Bharat means building systems Bharat can trust. Meesho Pragati Agent is designed, from its architecture upward, to be one.*

27. References

The following sources informed the research and claims made throughout this document. Figures and claims are paraphrased from these sources; readers are encouraged to consult the original sources directly for full context.

- Government of India, Press Information Bureau (PIB) — release on the estimated potential of AI-driven alternative-data lending models to unlock a portion of India's MSME credit gap (estimated USD 130–170 billion).
- Public reporting and regulatory filings concerning Meesho Payments Private Limited (MPPL), including its rights-issue capital infusion (~₹100 crore), year-on-year revenue growth, and reported net loss, in the context of its role as a Lending Service Provider partnering with third-party NBFCs.
- Industry analysis on MSME lending in India, addressing the prevalence of cash-based transactions among small businesses and the resulting difficulty traditional lenders face in verifying applicant cash flow without GST filings or bank-verified income statements.
- Industry analysis on the economics of small-ticket loan underwriting, addressing the finding that manual review cost for small loans is comparable to that for larger loans, disincentivizing profitable manual service of small borrowers.
- Amazon Web Services (AWS) — published EC2 on-demand pricing documentation, used to ground GPU inference hosting cost estimates (Section 21).
- Twilio — published WhatsApp Business API pricing documentation, and third-party analysis of Meta's 2026 WhatsApp per-message/per-conversation rate card for the Indian market, used to ground messaging cost estimates (Section 21).
- Reserve Bank of India — Digital Lending Guidelines, referenced in the context of deterministic, auditable credit-decision requirements (Sections 8, 11, 25).
- Government of India — Digital Personal Data Protection (DPDP) Act, referenced in the context of consent requirements for alternative-data underwriting (Section 25.4).